Devita Syifa
1A TTE

No.
Date:

Latihan Konsep Perubahan - Derivatif / Diferensial

1. $y = f(x) = -2x^2 + 18 \; ; \; x \geq 0$

2. Tentukan perubahan y dibagi perubahan x $\left(\frac{\Delta y}{\Delta x}\right)$ antara $X_0 = 2$ dan $X_1 = 2{,}005$

$X_0 = 2 \qquad X_1 = 2{,}005$

$y = f(x)$

$$\frac{\Delta y}{dx} = \frac{f(2{,}005) - f(2)}{2{,}005 - 2} = \frac{9{,}95 - 10}{0{,}005} = \frac{-0{,}05}{0{,}005} = -10$$

$$\frac{dy}{dx} = -4x$$

b. Tentukan $f'(x) = \frac{dy}{dx}$ di $x = 2$

$f(2) = -2(2)^2 + 18 = 10$

$$f'(2) = \lim_{x \to 2} \frac{-2x^2 + 18 - 10}{x - 2} = \lim_{x \to 2} \frac{-2x^2 + 8}{x - 2} = \lim_{x \to 2} \frac{-2(x^2 + 4)}{x - 2}$$

$$= \lim_{x \to 2} \frac{-2(x-2)(x+2)}{x - 2} = \lim_{x \to 2} -2(x+2) = -2(2+2) = -8$$

c. Tentukan gradien antara $x_0 = 2$ dan $x_1 = 2{,}005$

$\Delta x = 2{,}005 - 2 = 0{,}005$

$\Delta y = y_2 - y_1 = f(x_1) - f(x_0) = f(2{,}005) = f(2)$

$= 9{,}95 - 10 = -0{,}05$

$$\frac{\Delta y}{\Delta x} = \frac{-0{,}05}{0{,}005} = -10$$

d. Tentukan gradien singgung di titik $x = 2$ dan grafik !

- $m = f'(2) = -4 \cdot 2 = -8$
- Pers garis singgung

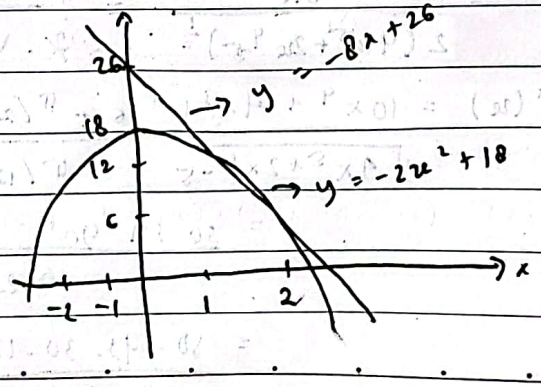
$(y - y_1) = m(x - x_1)$

$(y - 10) = -8(x - 2)$

$y - 10 = -8x - 16$

$y = -8x + 16 + 10$

$y = -8x + 26$



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Date:

2. Posisi benda memenuhi persamaan $s(t) = -t^2 + 4$ cm $(t > 0)$

a. Hitung $\frac{\Delta s}{\Delta t} = \frac{s(t+\Delta t) - s(t)}{\Delta t}$; kecepatan rata-rata !

$$\frac{\Delta s}{\Delta t} = \frac{s(t+\Delta t) - s(t)}{\Delta t}$$

$$= \frac{-(t+\Delta t)^2 + 4 - (-t^2 + 4)}{\Delta t}$$

$$= \frac{(t^2 + 2 + \Delta t + \Delta t^2) + 4 + t^2 - 4}{\Delta t}$$

$$= \frac{-t^2 - 2 + \Delta t - \Delta t^2 + t^2}{\Delta t}$$

$$= \frac{-2t + \Delta t - \Delta t^2}{\Delta t}$$

$$= \frac{(\Delta t - 2t - \Delta t)}{\Delta t} = -2t - \Delta t$$

b. Hitung kecepatan sesaat (yaitu jika Δt mendekati nol ; $\lim_{\Delta t \to 0} \frac{\Delta s}{\Delta t} = \frac{ds}{dt}$ turunan pertama dari s

$$\lim_{\Delta t \to 0} \frac{ds}{dt} = \lim_{\Delta t \to 0} -2t$$

3. Tentukan turunan pertama dari fungsi berikut

a. $\Delta t = t_2 - t_1$

$$= \frac{\pi}{4} - \frac{\pi}{6} = \frac{3\pi - 2\pi}{12}$$

$$\Delta t = \frac{\pi}{12} = 0,26167$$

$$i(t) = V(t) = \frac{\Delta s}{\Delta t} = \frac{s(t+\Delta t) - s(t)}{\Delta t}$$

$$= \frac{s(\pi/6 + \pi/2) - s(\pi/6)}{\pi/12}$$

b.

$$= \frac{50 \sin 90^\circ - 50 \sin 60^\circ}{0,26167}$$

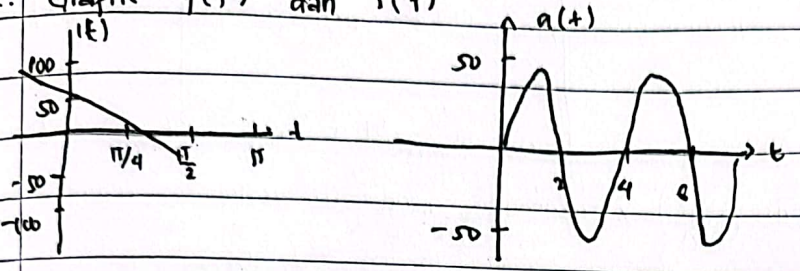
$$= \frac{50 \cdot 45 \cdot 30 \cdot 127}{0,26167} = 25.599 \text{ A}$$

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No.

Date:

c. Grafik $q(t)$ dan $i(t)$



4. $S(t) = -t^2 + 4t$

a. $S = 0$

$-t^2 + 4t = 0$

$-t(t-4)$

$-t = 0 \quad t = 4.$

b. $V = 0$

$S'(t) = V(t) = -2t + 4$

$V(1) = -2t + 4$

$-2t - 4 \geq 0$

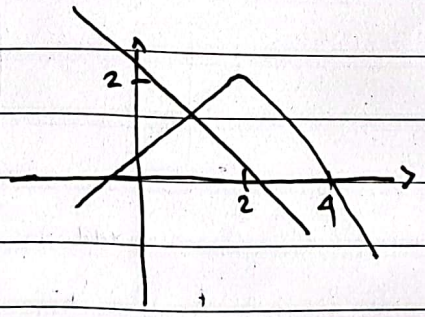
$-2t = -4$

$t = 2$

Positif saat $t \geq 2$ Negatif saat $t \leq 2$

c $S(t) = -t^2 + 4t$

$V(t) = -2t + t$



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